

Operating Windows and Thermal Averaging in a Fixed Graphene Strip Model

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Abstract—A successor to the finite-contact transmission audit retains its fitted width and tests interior detuning, incident energy, open-mode amplitudes and finite-temperature conductance weighting. Exact transfer identities provide an arbitrary-detuning amplitude. Directed interval calculations verify sufficient operating windows for a declared one-percent lowest-mode probability tolerance and a 0.01-radian relative phase tolerance. Thermal averaging at four declared temperatures changes lowest-mode transmission by at most approximately 0.0191 percent at 4 K. These are conditional continuum-model results, not graphene capacity enforcement, quantum-channel certification or physical latch validation.

Index Terms—Dirac transport, finite contacts, interval arithmetic, thermal averaging, MQM interfaces.

I. FROZEN SUCCESSOR PROTOCOL

Retain infinite-mass transverse boundaries, identical uniform-width leads and abrupt mode-preserving interfaces from the preceding MQM contact audit. The ideal strip background is described in [1]. Fix $L = 30$ nm, $\hbar v_F = 0.6582119$ eV nm and lead doping $\mu = 1/5$ eV. The previous fitted binary64 width $W \approx 15.795235$ nm is retained without refitting. Its exact binary64 rational value is used for interval calculations; this does not enclose fabrication uncertainty.

Let E be incident energy relative to the global Fermi level and U the uniform interior potential shift. Lead energy is $\mu + E$, while interior energy is $E - U$. Thus energy variation and gate detuning are distinct experiments. Freeze 401-point scans across $[-20, 20]$ meV for each, temperatures 0.02, 0.1, 1, 4 K, and a thermal integration window of $40k_B T$ on either side of zero. Disorder, interactions, magnetic fields and phonon scattering are absent. All results omit spin and valley multiplicity; conversion to conductance needs its own degeneracy contract.

The selected $T_* = 7/768$ is a per-mode engineering target. The acceptance band is relative lowest-mode probability error below $1/100$, together with relative transmitted phase below $1/100$ radians. These thresholds are frozen before successor evaluation. They are not the complete-channel 2^{-16} target. Neither the selected rational nor a probability tolerance represents an occupancy trigger or a Lindblad rate.

II. ARBITRARY-DETUNING TRANSFER AMPLITUDE

For mode $q = (n + 1/2)\pi/W$, set $p = (E - U)/(\hbar v_F)$ and $f = (\mu + E)/(\hbar v_F)$. An incoming channel exists only

for $f > q$, with $k_x = \sqrt{f^2 - q^2}$. The interior Dirac equation gives the transfer generator and matrix

$$A = q\sigma_z + ip\sigma_x, \quad A^2 = (q^2 - p^2)I, \quad (1)$$

$$P = cI + sA, \quad c = \cosh(\lambda L), \quad s = \frac{\sinh(\lambda L)}{\lambda}, \quad (2)$$

$$\lambda^2 = q^2 - p^2. \quad (3)$$

The continuous limit is $s = L$ at $\lambda = 0$. Imaginary λ gives the propagating-strip continuation. Current conservation follows from $A^\dagger \sigma_x + \sigma_x A = 0$ and $\det P = 1$.

Using the identical-lead spinors $u_\pm = (1, (\pm k_x + iq)/f)^T$, matching $P(u_+ + ru_-) = tu_+$ yields

$$t_n(E, U) = \left[c + i \frac{q^2 - fp}{k_x} s \right]^{-1}. \quad (4)$$

Then $T_n = |t_n|^2$; the phase convention uses fixed interface planes. At $p = 0$, this recovers the preceding finite-contact formula. Five exact symbolic checks verify the generator, current identity, determinant, denominator and neutral reduction. A total of 3285 binary64 checks verify direct matrix-exponential matching, flux conservation, sampled windows, the nominal value and quadrature comparisons. These are scoped checks, not independent review.

III. NUMERICAL BOUNDARIES AND INTERVAL CERTIFICATE

The nearest numerical crossings of the joint probability/phase band are approximately $[-0.621874, 0.620163]$ meV for incident energy at $U = 0$, and $[-0.694291, 0.696739]$ meV for detuning at $E = 0$. These roots and sampled interiors are numerical evidence, not rigorous maximal-window claims.

After observing the roots, smaller symmetric windows were selected as a disclosed engineering refinement. A separate directed interval calculation with 50 decimal digits verifies every box in Table I. For the lowest mode write the denominator as $D = c + ib$, and the nominal denominator as $D_0 = c_0 + ib_0$. Entire boxes must satisfy

$$\frac{99}{100}T_* < \frac{1}{c^2 + b^2} < \frac{101}{100}T_*, \quad (5)$$

$$R = c_0c + b_0b > 0, \quad (6)$$

$$|b_0c - c_0b| < \tan(1/100)R. \quad (7)$$

The latter inequalities bound $\arg(t/t_0)$ without averaging amplitudes. Exact rational source coefficients and an interval enclosure of π enter the calculation. Strict comparisons pass throughout the declared boxes; this is a computational interval certificate under the frozen equations. The independent joint

TABLE I
SUFFICIENT INTERVAL-VERIFIED OPERATING WINDOWS

Variation	Radius (eV)	Interval boxes
Energy only	1/2000	1000
Gate detuning only	1/2000	1000
Energy and gate, independently	1/4000 each	1600

TABLE II
THERMAL CHANGE RELATIVE TO SELECTED LOWEST-MODE TARGET

T (K)	Lowest mode (%)	All-mode sum (%)
0.02	$4.7770 \cdot 10^{-7}$	$2.7015 \cdot 10^{-5}$
0.10	$1.1943 \cdot 10^{-5}$	$3.8480 \cdot 10^{-5}$
1.00	0.0011943	0.0012208
4.00	0.0191166	0.0191431

rectangle is $|E|, |U| \leq 0.25 \text{ meV}$. One must not combine the larger separate windows into an untested joint rectangle. Interval probability bounds stay below approximately 0.568%, 0.509% and 0.539% respectively. No coefficient uncertainty or material variation is included. Selection after exploratory evaluation is disclosed, rather than presented as an independent prediction.

IV. ALL MODES AND THERMAL AVERAGING

At nominal energy, two modes are open. The second-mode threshold is $E \approx -3.627454 \text{ meV}$; incident-energy scans therefore change channel count. The replay stores probabilities and phases of every open mode. A closed channel has no transmitted phase and is excluded, rather than assigned a physical phase of zero.

Finite-temperature linear-response transport uses Fermi-derivative weighting, as in standard Landauer calculations [2]:

$$\bar{T}(T) = \int_{-\infty}^{\infty} \frac{\sum_{n \text{ open}} T_n(E, 0)}{4k_B T \cosh^2(E/(2k_B T))} dE. \quad (8)$$

This is a conductance weight, not a coherent quantum-channel average. Adaptive quadrature is compared with independent 256- and 512-node Gauss–Legendre integration on intervals

split at mode thresholds. All method differences are below $4.7 \cdot 10^{-17}$. Their agreement is numerical evidence; reported adaptive errors are estimates, not interval integration certificates. The total includes the already-present second-mode contribution and must not be called the lowest-mode probability. For $a = W/(\pi \hbar v_F)$, $C = a\mu + 1$, mode counting and $T_n \leq 1$ give the omitted-tail bound

$$\delta_{\text{tail}} \leq 2e^{-40}[C + 41ak_B T]. \quad (9)$$

It is below $2.24 \cdot 10^{-17}$ at the declared temperatures. This analytic tail estimate does not certify the interior quadrature or bound temperature-dependent scattering omitted from the model.

V. SCOPED RESULT AND REMAINING BRIDGE

The fixed continuum model supports a sufficient joint detuning/energy window and small thermal conductance corrections over the declared temperature range. No gate-voltage tolerance follows without electrostatic calibration. Contact boundary realization, disorder, material variation and the map from transport to a completion-conditioned reservoir remain OPEN. Physical promotion is zero. No hardware result, quantum advantage, error-correction threshold or autonomous native admission closure follows.

The next high-value gate is a typed transduction specification: define which observable signals completion, how it drives the physical reservoir, and what output information the reservoir can distinguish. In parallel, test an explicit interface model with reverse jumps and finite reservoir bandwidth. Transport robustness and quantum-information preservation require separate acceptance criteria.

REFERENCES

- [1] J. Tworzydło et al., “Sub-Poissonian Shot Noise in Graphene,” *Phys. Rev. Lett.*, vol. 96, 246802, 2006.
- [2] “Enhancing the thermoelectric figure of merit in engineered graphene nanoribbons,” *Beilstein J. Nanotechnol.*, vol. 6, 2015, article 119. <https://www.beilstein-journals.org/bjnano/articles/6/119>.